Implicit Differentiation.

Definition Explicit VS Implicit

Explicit function : y is expressed directly in terms of x e.g. $y = ax^2 + bx + c$

Implicit function : x and y are mixed together. y is not isolated e.g. $x^2 + y^2 = 25$

General form : $F(x, y) = 0$

Example : $x^2 + y^2 = 1$ (for positive case)

① Explicit method

$$y^2 = 1 - x^2$$

$$y = \sqrt{1-x^2} = (1-x^2)^{\frac{1}{2}}$$

$$y' = \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot (-2x)$$

$$= \frac{-x}{\sqrt{1-x^2}}$$

② Implicit method

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(1).$$

$$2x + 2y \cdot \frac{dy}{dx} = 0.$$

chain rule

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y} = \frac{-x}{\sqrt{1-x^2}}$$

Inverse function

$$f(x) = y \Rightarrow f^{-1}(y) = x$$

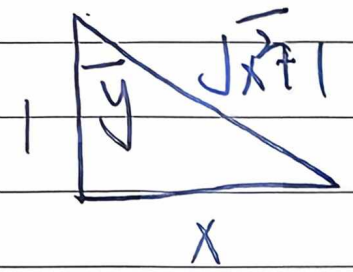
A function must be one-to-one (injective)

Example $y = \arctan(x)$.

$$\tan y = x \qquad \frac{dy}{dx} = \cos^2 y.$$

$$\frac{d}{dx}(\tan y) \cdot \frac{dy}{dx} = \frac{d}{dx} x$$

$$\frac{1}{\cos^2 y} \cdot \frac{dy}{dx} = 1$$



$$\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2} \Leftarrow \cos y = \frac{1}{\sqrt{x^2+1}}$$

$$g(x) = f^{-1}(x).$$

$$f(g(x)) = x.$$

$$\frac{d}{dx} f(g(x)) = \frac{d}{dx} x \Rightarrow \frac{d}{dx}[f^{-1}(x)] = \frac{1}{f'(f^{-1}(x))}$$

$$f'(g(x)) \cdot g'(x) = 1$$

$$g'(x) = \frac{1}{f'(g(x))}$$

$$\frac{d}{dx}(e^x) = e^x \quad \frac{d}{dx}(\ln x) = \frac{1}{x} \quad \frac{d}{dx}(a^x) = \ln a \cdot a^x$$

Find $\frac{d}{dx}(a^x)$

$$\frac{d}{dx}(a^x) = \lim_{x \to 0} \frac{a^{x+\Delta x} - a^x}{\Delta x} = a^x \lim_{x \to 0} \frac{a^{\Delta x} - 1}{\Delta x}$$

$$M(a) = \lim_{x \to 0} \frac{a^{\Delta x} - 1}{\Delta x} \qquad \frac{d}{dx}(a^x) = M(a)\, a^x$$

$M(a)$ is the slope of the graph $y = a^x$ at $a = 0$

Method ① write e and use chain rule

rewrite $a = e^{\ln a}$ $\quad a^x = (e^{\ln a})^x = e^{x \ln a}$

$$\frac{d}{dx}(a^x) = \frac{d}{dx}(e^{x \ln a}) = \frac{d}{dx}(x \ln a) \cdot \frac{d}{dy}(e^y)$$

$y = x \ln a$

$$= \ln a \cdot e^{x \ln a}$$

$$= \ln a \cdot a^x.$$

$$M(a) = \ln a \quad \frac{d}{dx}(a^x) = M(a) \cdot a^x = \ln a \cdot a^x$$

Method ② Logarithmic Differentiation

to find $\frac{d}{dx} f(x)$ by finding $\frac{d}{dx} \ln(f(x))$ instead

$$y = f(x) = a^x$$

$$\ln(y) = \ln(a^x) = x \ln a$$

$$\frac{dy}{dx} \cdot \frac{1}{y} = (1 \cdot \ln a)$$

$$\frac{dy}{dx} \cdot \frac{1}{y} = \ln a$$

$$\frac{dy}{dx} = y \cdot \ln a = a^x \cdot \ln a$$

Example ① $\frac{d}{dx}(x^x)$

$$y = x^x$$

$$\ln y = \ln x^x = x \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{d}{dx}(x \ln x) = \ln x + x \cdot \frac{1}{x} = \ln x + 1.$$

$$\frac{dy}{dx} = y \cdot (\ln x + 1) = x^x \cdot (\ln x + 1)$$

Example ② evalvate $\lim_{n\to\infty} (1+\frac{1}{n})^n$

$$\ln[(1+\frac{1}{n})^n] = n\cdot\ln(1+\frac{1}{n})$$

$\Delta x = \frac{1}{n}$ $\quad n\to\infty \quad \Delta x \to 0.$

$$= \frac{1}{\Delta x}\ln(1+\Delta x)$$

$$= \frac{\ln(1+\Delta x) - \ln(1)}{\Delta x} \quad (\ln(1)=0)$$

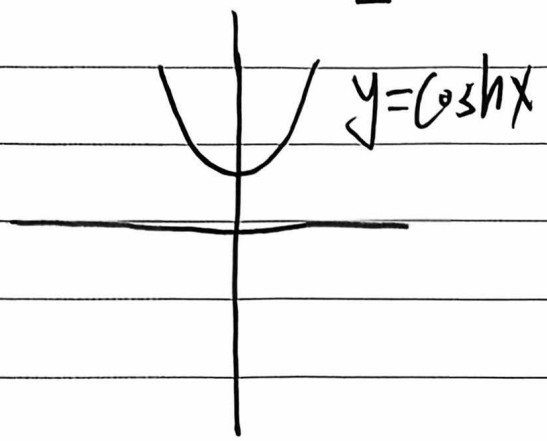
$f(x) = \ln x \quad = f'(x)$ at $x=1 = \frac{d}{dx}\ln x\Big|_{x=1}$

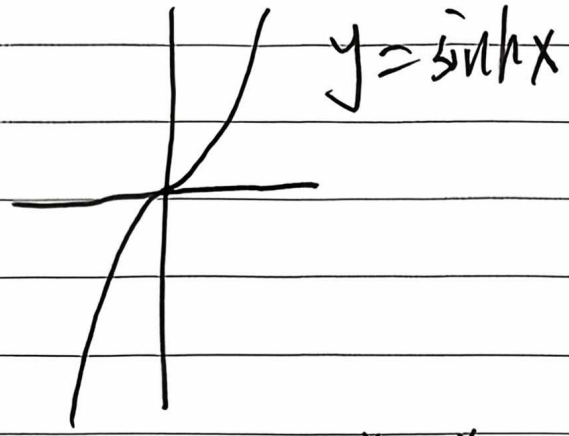
$$\lim_{n\to\infty} \ln[(1+\frac{1}{n})^n] = 1$$

$$\lim_{n\to\infty} (1+\frac{1}{n})^n = e^1 = e$$

Hyperbolic trig functions

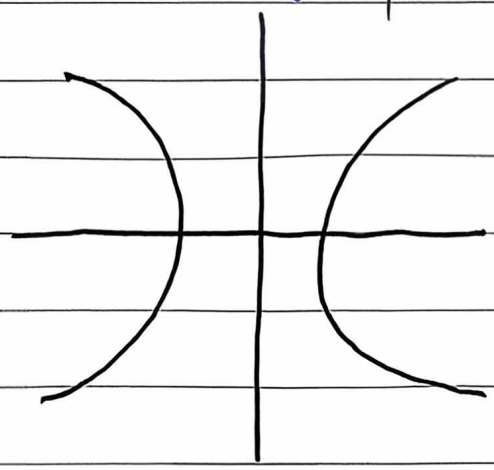
$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \sinh x = \frac{e^x - e^{-x}}{2}$$





$$\frac{d}{dx}\cosh x = \frac{e^x - e^{-x}}{2} = \sinh x \qquad \frac{d}{dx}\sinh x = \frac{e^x + e^{-x}}{2} = \cosh x$$

$x^2 - y^2 = 1$



$x = \cosh x$

$y = \sinh x$

$\cosh^2 x - \sinh^2 x = 1$

$= \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$

$= 1$

$$\frac{d}{dx} x^r = r \cdot x^{r-1}$$

Method ① (base e)

$$x = e^{\ln x}$$

$$x^r = (e^{\ln x})^r = e^{r \cdot \ln x}$$

$$\frac{d}{dx} \cdot x^r = \frac{d}{dx}(e^{r \cdot \ln x})$$

$$= \frac{r}{x} \cdot e^{r \ln x}$$

$$= \frac{r}{x} \cdot x^r = r \cdot x^{r-1}$$

Method ② (log diff)

$$y = x^r$$

$$\ln y = \ln x^r = r \cdot \ln x$$

$$\frac{dy}{dx} \cdot \frac{1}{y} = \frac{r}{x}$$

$$\frac{dy}{dx} = \frac{r}{x} \cdot y = \frac{r}{x} \cdot x^r = \cancel{r x^n}\ r \cdot x^{r-1}$$

$$\frac{y'}{y} = (\ln y)'$$

$$y' = y \cdot (\ln y)'$$